Chaitanya Mamatha Ananda

Compiler Optimization Engineer · Ph.D. Candidate · LLVM/BOLT Post-Link Optimization

Riverside, CA · 951-907-8519 · cmama002@ucr.edu · LinkedIn · Google Scholar

Summary

Compiler Ph.D. candidate specializing in post-link and profile-guided optimization for x86 and Arm (AArch64). First-author on **CGO**, **LCTES**, and **CAIS/MLArchSys** publications: heap-layout optimization that reduces execution time by **21.7%** on average, and cross-module basic block deduplication that shrinks binaries by up to **25.8%**. Recently at Google, applied AI-driven search (AlphaEvolve) to warehouse-scale interprocedural code layout.

Technical Skills

Languages & Tools: C++, C, Python, NumPy, Linux, Git

Compilers & Optimization: LLVM, BOLT, Propeller, profile-guided optimization (PGO/FDO), post-link and binary optimization, code and memory layout, basic block reordering, interprocedural optimization

Performance Engineering: x86 and Arm (AArch64), hardware performance counters, perf, profiling, benchmarking, warehouse-scale workloads

ML for Compilers: AI-driven heuristic search, AlphaEvolve, Vizier, Evolution Strategies, MLGO

Experience

Google - Student Researcher — **Sunnyvale, CA · Sep 2025 – May 2026**

AI-driven code layout optimization for warehouse-scale binaries; Manager: Sriraman Tallam

- Extended Google's **Propeller** post-link optimizer with AI-driven interprocedural code layout for warehouse-scale binaries, guided by hardware performance counters.
- Improved execution time by **1.6%** on *LLVM Clang* and **0.23%** on a production *Search* workload, where even marginal gains compound to substantial fleet-wide savings.

University of California, Riverside - Ph.D. Researcher — **Riverside, CA · Sep 2022 – Present**

Data locality improvement and code footprint reduction; Advisor: Dr. Rajiv Gupta

- **PreFix:** Profiling-guided heap optimization that groups hot objects to improve locality, reducing execution time by **21.7%** on average.
- **DeduBB:** Propeller-based post-link optimizer for x86 and Arm that deduplicates basic blocks across module boundaries, reducing binary size by up to **25.8%** with no performance loss under PGO.

Indian Institute of Science - Undergraduate Researcher — **Bengaluru, India · Aug 2019 – May 2022**

Parallel programming and scientific data analysis

- Built a parallel programming model for PDE solvers using **Regent/Legion**.
- Designed an anomaly detector for scientific data using statistical and deep-learning methods.

Publications

- **CGO 2025**: *PreFix: Optimizing the Performance of Heap-Intensive Applications.* (DOI · Google)
  C. Mamatha Ananda, R. Gupta, S. Tallam, H. Shen, and X.D. Li.
- **LCTES 2026**: *DeduBB: Binary Code Size Reduction via Post-Link Basic Block Deduplication.* (DOI · Google)
  C. Mamatha Ananda, M. Afarin, R. Gupta, S. Tallam, H. Shen, and X.D. Li.
- **CAIS 2026 / MLArchSys 2026**: *AI-PROPELLER: Warehouse-Scale Interprocedural Code Layout Optimization with AlphaEvolve.* (arXiv · Google)
  C. Mamatha Ananda, R. Gupta, M. Trofin, A. Grossman, S. Tallam, X.D. Li, and A. Yazdanbakhsh.

Education

Ph.D. in Computer Science, UC Riverside — **Sep 2022 - Present**

B.E. in Computer Science and Engineering, Bangalore Institute of Technology, India — **2021**

Awards & Teaching

Award: Dean's Distinguished Fellowship, UC Riverside (2022) · **Teaching:** Compiler Design and Construction.